Administrarea tabelelor
(Cap. 11)

1) Crearea unei tabele permanente:

```
SQL> CREATE TABLE studenti
        (facultate char(30)  DEFAULT  'Automatica si Calculatoare',
         catedra char(20),
        cnp number(13),
        nume     varchar2(30),
        data_nastere date,
        an_univ number(4)  DEFAULT  2006,
         media_admitere number(5,2) ,
        discip_oblig  varchar2(20) DEFAULT 'Matematica',
        discip_opt     varchar2(20) DEFAULT 'Fizica' ,
        operator  varchar2(20) DEFAULT  user,
        data_op   date DEFAULT sysdate )
        TABLESPACE ubd_data,
        STORAGE (initial 8M),
        PCTUSED 30,
        PCTFREE 20,
        NOLOGGING;
```

2) Crearea unei tabele permanente partitionate:

```
CREATE TABLE angajati_dep
 (
 depart number(2),
 ecuson number(5),
 nume varchar2(30),
 job  varchar2(20),
 data_ang date,
 salariu number(6)
 )
PARTITION BY LIST (depart)
 (
 PARTITION Dep10 VALUES (10),
 PARTITION Dep20 VALUES (20),
 PARTITION Dep30 VALUES (30)
 );

SQL> insert into angajati_dep select deptno, empno, ename, job, hiredate, sal from emp;
```

SQL> select * from angajati_dep partition (dep10);

DEPART	ECUSON	NUME	JOB	DATA_ANG	SALARIU
10	7782	CLARK	MANAGER	09-JUN-81	2450
10	7839	KING	PRESIDENT	17-NOV-81	5000
10	7934	MILLER	CLERK	23-JAN-82	1300

3) Crearea unei tabele temporare de catre userul curent:

a) SQL> create global temporary table emp_temp **on commit delete rows** as
select * from emp;
table created.

Observatii:

- Datele se pastreaza in tabela doar pe perioada tranzactiei, dar tabela ramane creata si in sesiunile urmatoare.

  SQL> select * from emp_temp;
  no rows selected

  SQL> insert into emp_temp select * from emp;
  14 rows created.

  SQL> select * from emp_temp;
  14 rows selected

  SQL>commit;

  SQL> select * from emp_temp;
  no rows selected

- Datele pot fi sterse din tabela iar tabela poate fi stearsa din dictionar in sesiunea curenta.

SQL> drop table emp_temp;
Table dropped.

b) SQL> create global temporary table emp_temp **on commit preserve rows** as
select * from emp;

Observatii:

- Datele se pastreaza in tabela doar pe perioada **sesiunii curente**, dar tabela ramane creata si in sesiunile urmatoare.

SQL> select count(*) from emp_temp;

COUNT(*)

14

- Datele pot fi sterse din tabela, dar tabela nu poate fi stearsa din dictionar in sesiunea curenta.

SQL> drop table emp_temp;

drop table emp_temp
*
ERROR at line 1:
ORA-14452: attempt to create, alter or drop an index on temporary table already in use

4) Vizualizarea ROWID-rilor pentru fiecare linie din tabela:

SQL> create table emp_copy as select * from scott.emp;

SQL> select rowid, empno, ename from emp_copy;

ROWID	EMPNO	ENAME
AAAWc2AAHAAAACXAAA	7369	SMITH
AAAWc2AAHAAAACXAAB	7499	ALLEN
AAAWc2AAHAAAACXAAC	7521	WARD
AAAWc2AAHAAAACXAAD	7566	JONES
AAAWc2AAHAAAACXAAE	7654	MARTIN
AAAWc2AAHAAAACXAAF	7698	BLAKE
AAAWc2AAHAAAACXAAG	7782	CLARK
AAAWc2AAHAAAACXAAH	7788	SCOTT
AAAWc2AAHAAAACXAAI	7839	KING
AAAWc2AAHAAAACXAAJ	7844	TURNER
AAAWc2AAHAAAACXAAK	7876	ADAMS
AAAWc2AAHAAAACXAAL	7900	JAMES
AAAWc2AAHAAAACXAAM	7902	FORD
AAAWc2AAHAAAACXAAN	7934	MILLER

5) Vizualizarea ROWNUM-rilor pentru fiecare linie din tabela:

SQL> select rownum, empno, ename from emp_copy where rownum<10;

ROWNUM	EMPNO	ENAME
1	7369	SMITH
2	7499	ALLEN
3	7521	WARD
4	7566	JONES
5	7654	MARTIN
6	7698	BLAKE
7	7782	CLARK
8	7788	SCOTT
9	7839	KING

6) Alocarea unei extensii la o tabela:

```
SQL> alter tablespace users
        add datafile 'e:/temp/ubd_data1.dbf'  size 10M;
Tablespace altered.

SQL>  alter table emp_copy
        allocate extent(size 500k datafile  'e:/temp/ubd_data1.dbf');
Table altered.
```

7) Stergerea unei coloane dintr-o tabela:

```
SQL>  alter table emp_copy
        drop column comm
        cascade constraints checkpoint 1000;
```

8) Redenumirea unei coloane dintr-o tablela:

```
SQL>  alter table emp_copy
        rename column sal to salary;
```

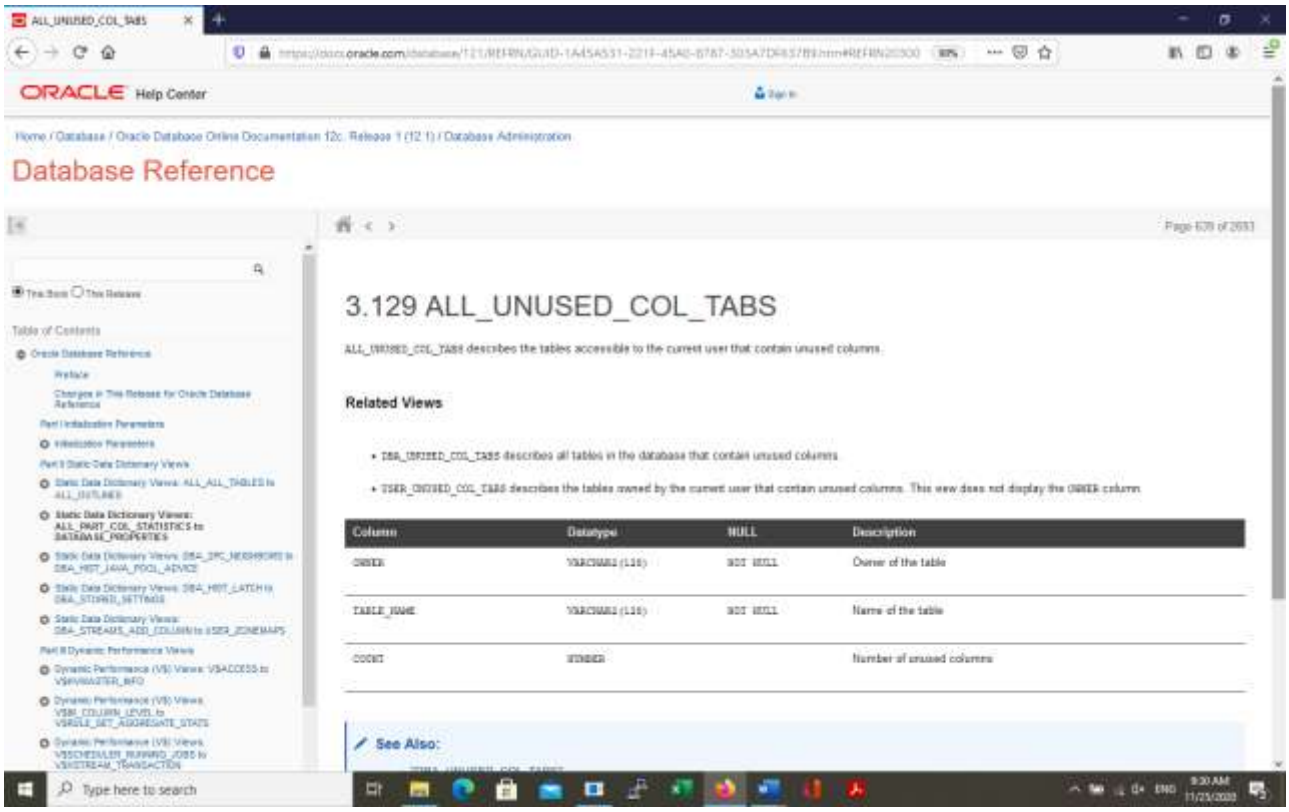
9) Dezactivarea unei coloane dintr-o tabela:

```
SQL>  alter table emp_copy
        set unused column sal
        cascade constraints;
```

10) Informatii despre tabele cu coloane dezactivate:

```
SQL> desc dba_unused_col_tabs;
```

Name	Null?	Type
OWNER	NOT NULL	VARCHAR2(30)
TABLE_NAME	NOT NULL	VARCHAR2(30)
COUNT		NUMBER



SQL> select * from dba_unused_col_tabs;

OWNER	TABLE_NAME	COUNT
ABD1	EMP_COPY	1

11) Stergerea din dictionar a coloanelor dezactivate dintr-o tabela:

```
SQL>  alter table emp_copy
        drop unused columns checkpoint 1000;

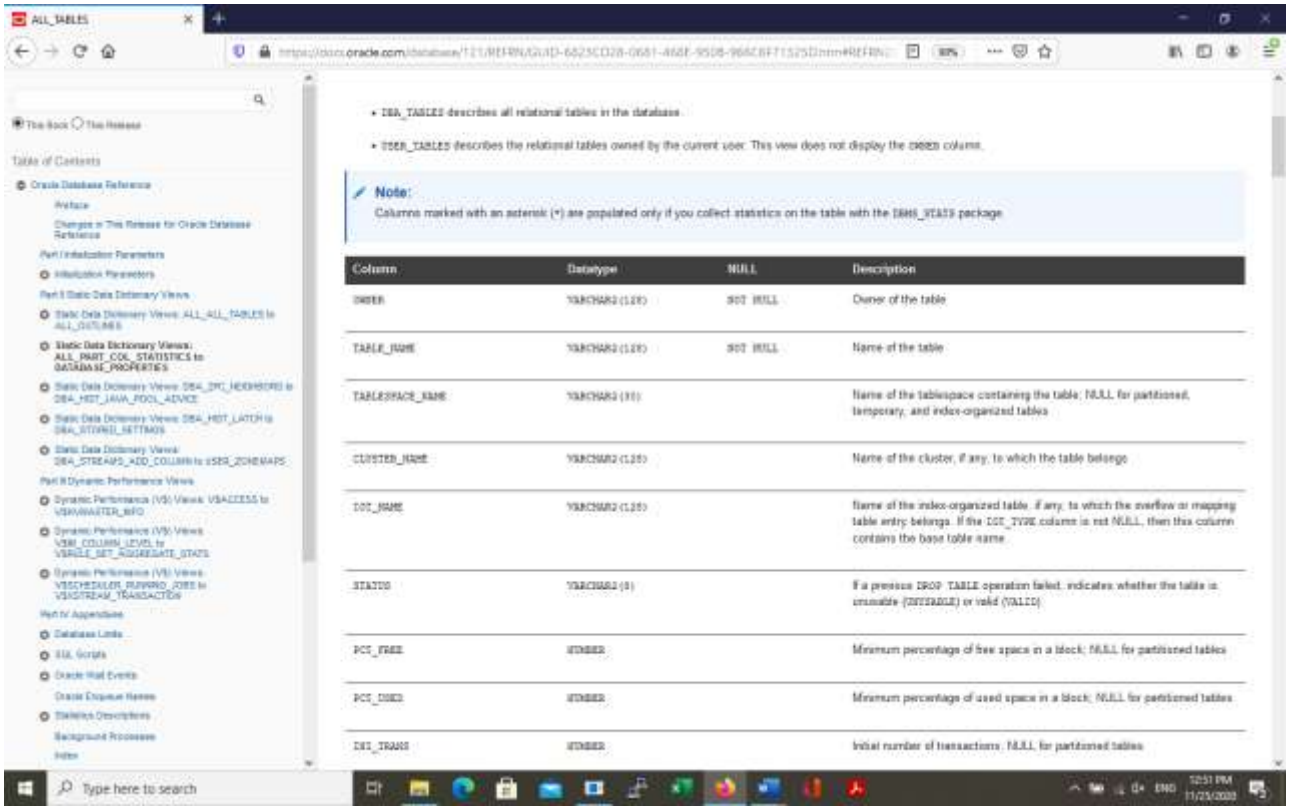
SQL> select * from dba_unused_col_tabs;
no rows selected
```

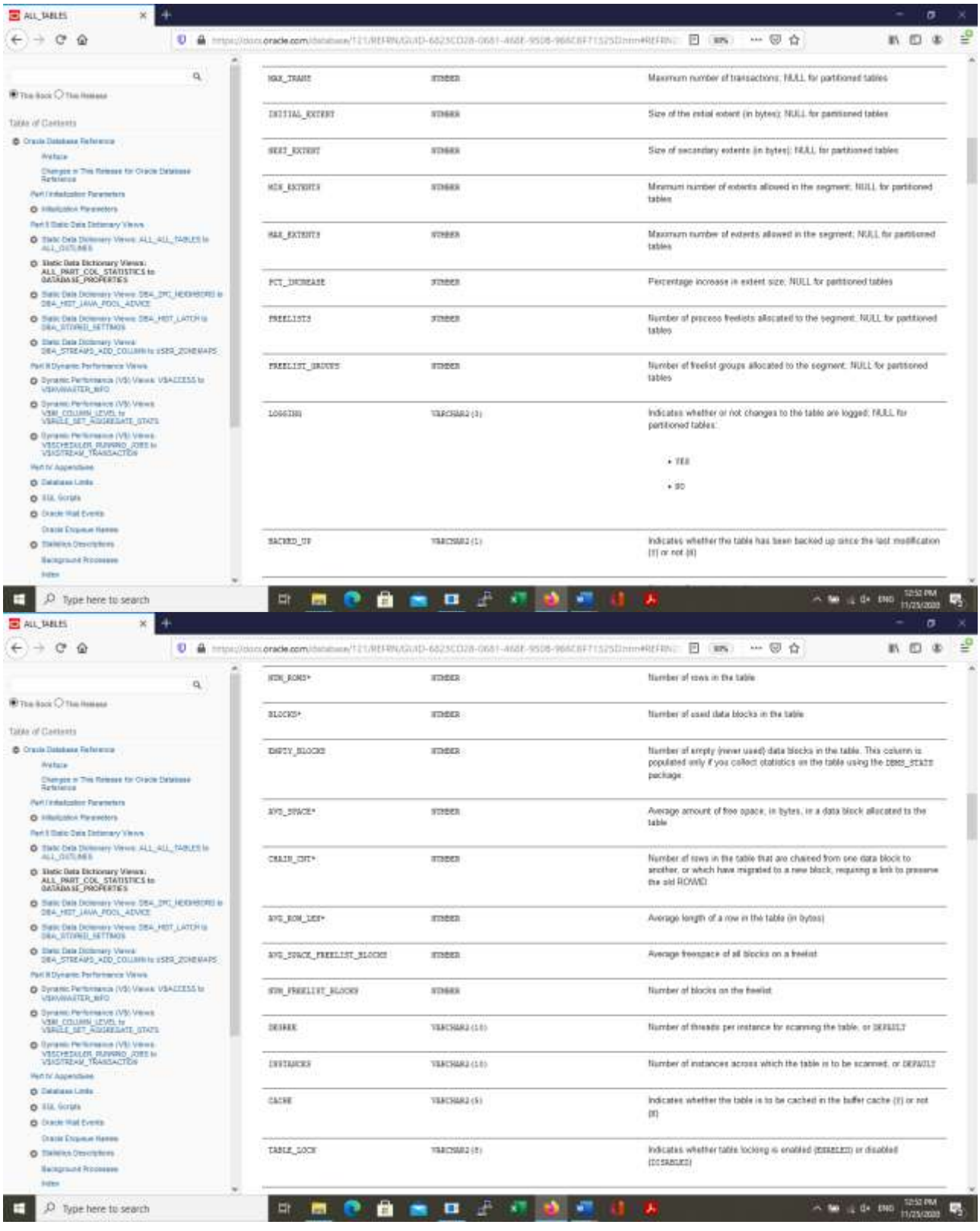
12) Informatii despre toate tabelele din baza de date :

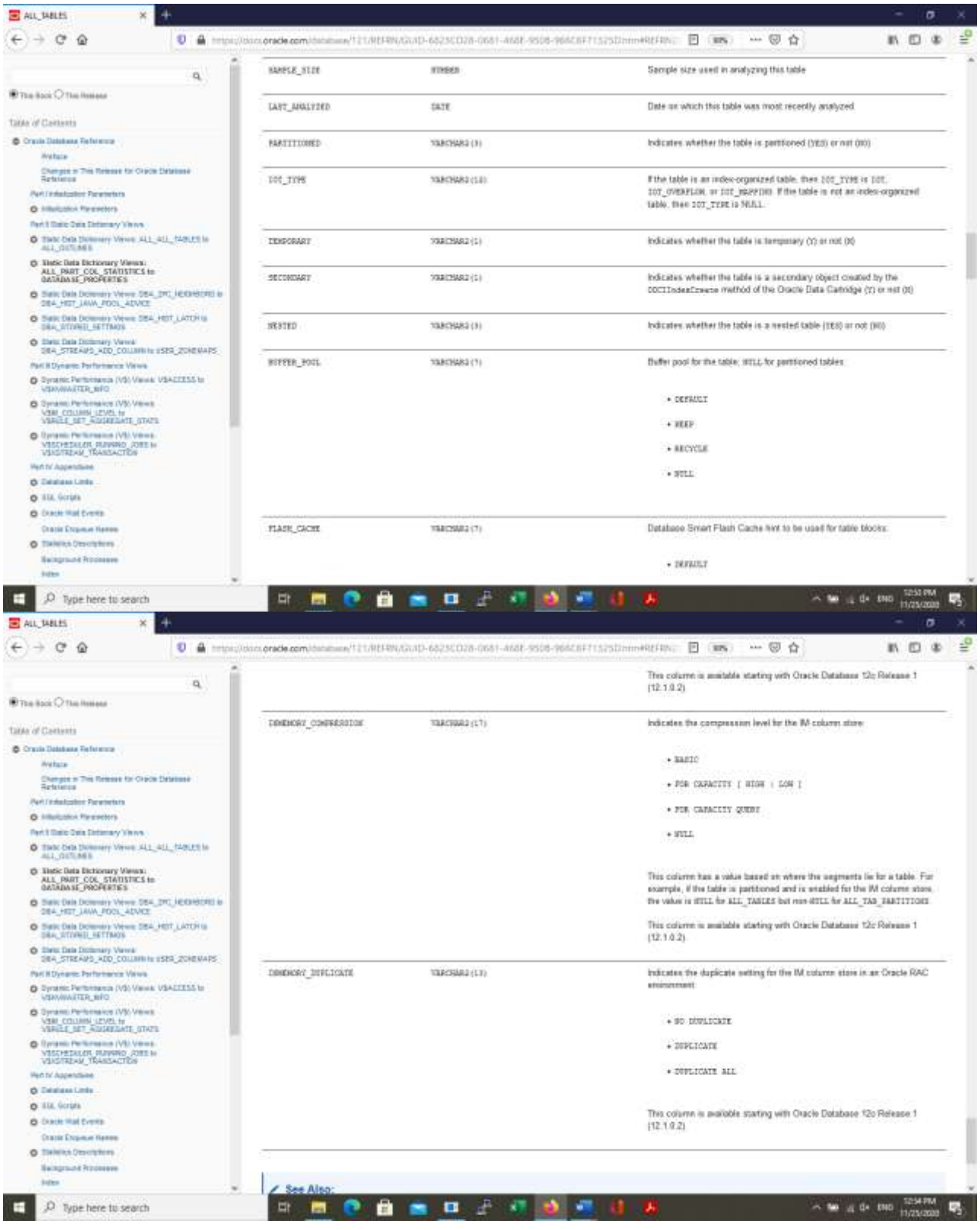
SQL> desc dba_tables;

Nume	Nul?	Type
OWNER	NOT NULL	VARCHAR2(30)
TABLE_NAME	NOT NULL	VARCHAR2(30)
TABLESPACE_NAME		VARCHAR2(30)
CLUSTER_NAME		VARCHAR2(30)
IOT_NAME		VARCHAR2(30)
PCT_FREE		NUMBER
PCT_USED		NUMBER
INI_TRANS		NUMBER
MAX_TRANS		NUMBER
INITIAL_EXTENT		NUMBER
NEXT_EXTENT		NUMBER
MIN_EXTENTS		NUMBER
MAX_EXTENTS		NUMBER
PCT_INCREASE		NUMBER
FREELISTS		NUMBER
FREELIST_GROUPS		NUMBER
LOGGING		VARCHAR2(3)
BACKED_UP		VARCHAR2(1)
NUM_ROWS		NUMBER
BLOCKS		NUMBER
EMPTY_BLOCKS		NUMBER
AVG_SPACE		NUMBER
CHAIN_CNT		NUMBER
AVG_ROW_LEN		NUMBER
AVG_SPACE_FREELIST_BLOCKS		NUMBER
NUM_FREELIST_BLOCKS		NUMBER
DEGREE		VARCHAR2(10)
INSTANCES		VARCHAR2(10)
CACHE		VARCHAR2(5)
TABLE_LOCK		VARCHAR2(8)
SAMPLE_SIZE		NUMBER
LAST_ANALYZED		DATE
PARTITIONED		VARCHAR2(3)
IOT_TYPE		VARCHAR2(12)
TEMPORARY		VARCHAR2(1)
SECONDARY		VARCHAR2(1)
NESTED		VARCHAR2(3)
BUFFER_POOL		VARCHAR2(7)
ROW_MOVEMENT		VARCHAR2(8)
GLOBAL_STATS		VARCHAR2(3)
USER_STATS		VARCHAR2(3)
DURATION		VARCHAR2(15)
SKIP_CORRUPT		VARCHAR2(8)
MONITORING		VARCHAR2(3)

CLUSTER_OWNER	VARCHAR2(30)
DEPENDENCIES	VARCHAR2(8)
COMPRESSION	VARCHAR2(8)
COMPRESS_FOR	VARCHAR2(30)
DROPPED	VARCHAR2(3)
READ_ONLY	VARCHAR2(3)
SEGMENT_CREATED	VARCHAR2(3)
RESULT_CACHE	VARCHAR2(7)
CLUSTERING	VARCHAR2(3)
ACTIVITY_TRACKING	VARCHAR2(23)
DML_TIMESTAMP	VARCHAR2(25)
HAS_IDENTITY	VARCHAR2(3)
CONTAINER_DATA	VARCHAR2(3)
INMEMORY	VARCHAR2(8)
INMEMORY_PRIORITY	VARCHAR2(8)
INMEMORY_DISTRIBUTE	VARCHAR2(15)
INMEMORY_COMPRESSION	VARCHAR2(17)
INMEMORY_DUPLICATE	VARCHAR2(13)




ALL_TABLES
Table of Contents
MAX_TRANS
INITIAL_EXTENT
NEXT_EXTENT
MIN_EXTENTS
MAX_EXTENTS
PCT_INCREASE
FREELISTS
FREELIST_GROUPS
LOGGING
BACKED_UP
NUM_ROWS*
BLOCKS*
EMPTY_BLOCKS
AVG_SPACE*
CHAIN_CNT*
AVG_ROW_LEN*
AVG_SPACE_FREELIST_BLOCKS
NUM_FREELIST_BLOCKS
DEGREE
INSTANCES
CACHE
TABLE_LOCK
Type here to search



SAMPLE_SIZE	NUMBER	Sample size used in analyzing this table
LAST_ANALYZED	DATE	Date on which this table was most recently analyzed
PARTITIONED	VARCHAR2 (3)	Indicates whether the table is partitioned (YES) or not (NO)
IOT_TYPE	VARCHAR2 (12)	If the table is an index-organized table, then IOT_TYPE is IOT, IOT_OVERFLOW, or IOT_MAPPING. If the table is not an index-organized table, then IOT_TYPE is NULL.
TEMPORARY	VARCHAR2 (1)	Indicates whether the table is temporary (Y) or not (N)
SECONDARY	VARCHAR2 (1)	Indicates whether the table is a secondary object created by the ODCIIndexCreate method of the Oracle Data Cartridge (Y) or not (N)
NESTED	VARCHAR2 (3)	Indicates whether the table is a nested table (YES) or not (NO)
BUFFER_POOL	VARCHAR2 (7)	Buffer pool for the table; NULL for partitioned tables • DEFAULT • KEEP • RECYCLE • NULL
FLASH_CACHE	VARCHAR2 (7)	Database Smart Flash Cache hint to be used for table blocks: • DEFAULT

		This column is available starting with Oracle Database 12c Release 1 (12.1.0.2)
INMEMORY_COMPRESSION	VARCHAR2 (17)	Indicates the compression level for the IM column store: • BASIC • FOR CAPACITY [HIGH | LOW] • FOR CAPACITY QUERY • NULL This column has a value based on where the segments lie for a table. For example, if the table is partitioned and is enabled for the IM column store, the value is USER for ALL_TABLES but non-USER for ALL_TAB_PARTITIONS This column is available starting with Oracle Database 12c Release 1 (12.1.0.2)
INMEMORY_DUPLICATE	VARCHAR2 (13)	Indicates the duplicate setting for the IM column store in an Oracle RAC environment: • NO DUPLICATE • DUPLICATE • DUPLICATE ALL This column is available starting with Oracle Database 12c Release 1 (12.1.0.2)

See Also:

```
SQL> select owner, tablespace_name, table_name  from dba_tables
       where owner = 'SCOTT';

OWNER                          TABLESPACE_NAME                TABLE_NAME
------------------------------ ------------------------------ ------------
SCOTT                                USERS                    CATALOG
SCOTT                                USERS                    DISCIPLINE
SCOTT                                USERS                    DEPT
SCOTT                                USERS                    EMP
SCOTT                                USERS                    BONUS
SCOTT                                USERS                    SALGRADE
```

13) Informatii despre tabelele create de userul curent :

```
SQL> desc user_tables

 Name                                      Null?                Type
 ----------------------------------------- --------             ------
 TABLE_NAME                                NOT NULL             VARCHAR2(30)
 TABLESPACE_NAME                                                VARCHAR2(30)
 CLUSTER_NAME                                                   VARCHAR2(30)
 IOT_NAME                                                       VARCHAR2(30)
 STATUS                                                         VARCHAR2(8)
 PCT_FREE                                                       NUMBER
 PCT_USED                                                       NUMBER
 INI_TRANS                                                      NUMBER
 MAX_TRANS                                                      NUMBER
 INITIAL_EXTENT                                                 NUMBER
 NEXT_EXTENT                                                    NUMBER
 MIN_EXTENTS                                                    NUMBER
 MAX_EXTENTS                                                    NUMBER
 PCT_INCREASE                                                   NUMBER
 FREELISTS                                                      NUMBER
 FREELIST_GROUPS                                                NUMBER
 LOGGING                                                        VARCHAR2(3)
 BACKED_UP                                                      VARCHAR2(1)
 NUM_ROWS                                                       NUMBER
 BLOCKS                                                         NUMBER
 EMPTY_BLOCKS                                                   NUMBER
 AVG_SPACE                                                      NUMBER
 CHAIN_CNT                                                      NUMBER
 AVG_ROW_LEN                                                    NUMBER
 AVG_SPACE_FREELIST_BLOCKS                                      NUMBER
 NUM_FREELIST_BLOCKS                                            NUMBER
 DEGREE                                                         VARCHAR2(10)
 INSTANCES                                                      VARCHAR2(10)
```

CACHE	VARCHAR2(5)
TABLE_LOCK	VARCHAR2(8)
SAMPLE_SIZE	NUMBER
LAST_ANALYZED	DATE
PARTITIONED	VARCHAR2(3)
IOT_TYPE	VARCHAR2(12)
TEMPORARY	VARCHAR2(1)
SECONDARY	VARCHAR2(1)
NESTED	VARCHAR2(3)
BUFFER_POOL	VARCHAR2(7)
ROW_MOVEMENT	VARCHAR2(8)
GLOBAL_STATS	VARCHAR2(3)
USER_STATS	VARCHAR2(3)
DURATION	VARCHAR2(15)
SKIP_CORRUPT	VARCHAR2(8)
MONITORING	VARCHAR2(3)
CLUSTER_OWNER	VARCHAR2(30)
DEPENDENCIES	VARCHAR2(8)
COMPRESSION	VARCHAR2(8)
DROPPED	VARCHAR2(3)
READ_ONLY	VARCHAR2(3)
SEGMENT_CREATED	VARCHAR2(3)
RESULT_CACHE	VARCHAR2(7)
CLUSTERING	VARCHAR2(3)
ACTIVITY_TRACKING	VARCHAR2(23)
DML_TIMESTAMP	VARCHAR2(25)
HAS_IDENTITY	VARCHAR2(3)
CONTAINER_DATA	VARCHAR2(3)
INMEMORY	VARCHAR2(8)
INMEMORY_PRIORITY	VARCHAR2(8)
INMEMORY_DISTRIBUTE	VARCHAR2(15)
INMEMORY_COMPRESSION	VARCHAR2(17)
INMEMORY_DUPLICATE	VARCHAR2(13)

SQL> select instances, tablespace_name, table_name from user_tables;

INSTANCES	TABLESPACE_NAME	TABLE_NAME
1	BD_DATA	ANGAJATI
1		ANGAJATI_DEP
1	BD_DATA	DEPT
1	BD_DATA	EMP
1		EMP_TEMP
1	BD_DATA	BONUS
1	BD_DATA	SALGRADE
1	BD_DATA	STUDENTI

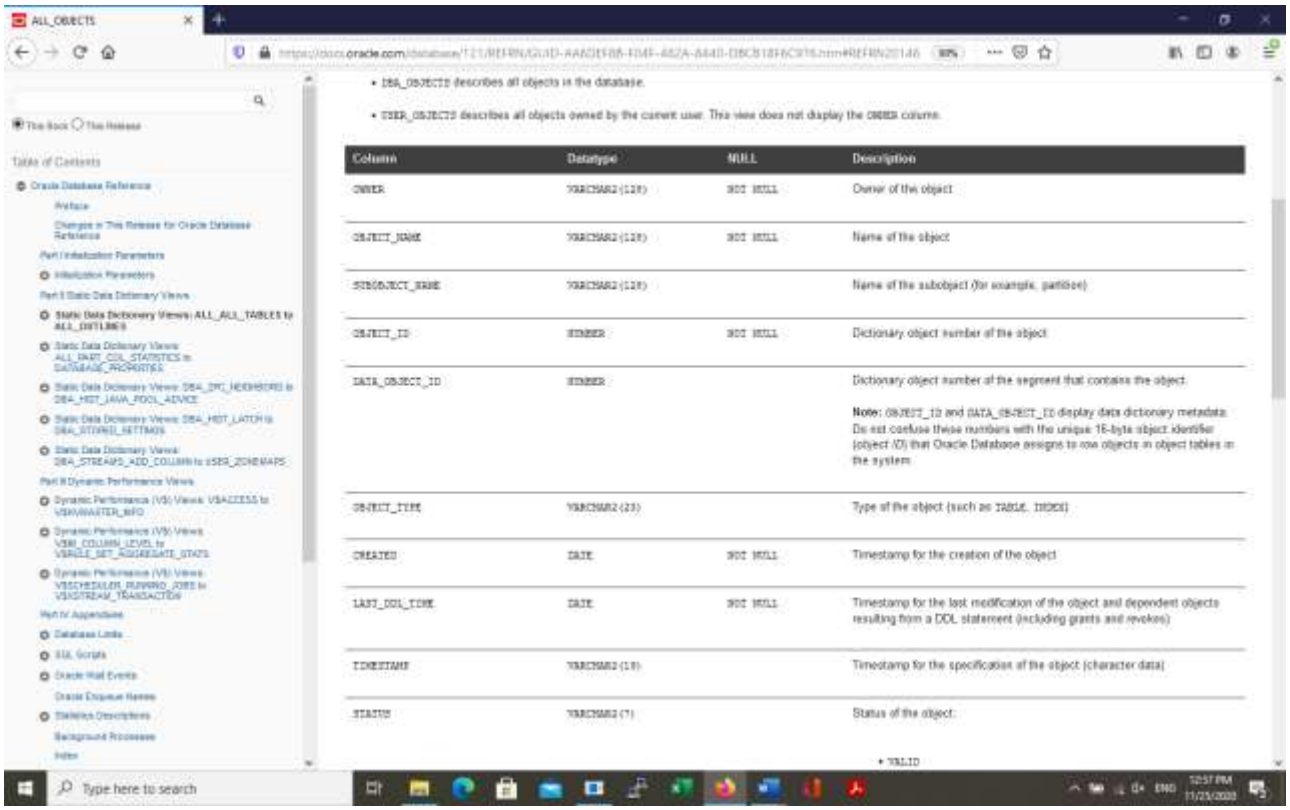
14) Informatii despre tabelele create de userul curent :

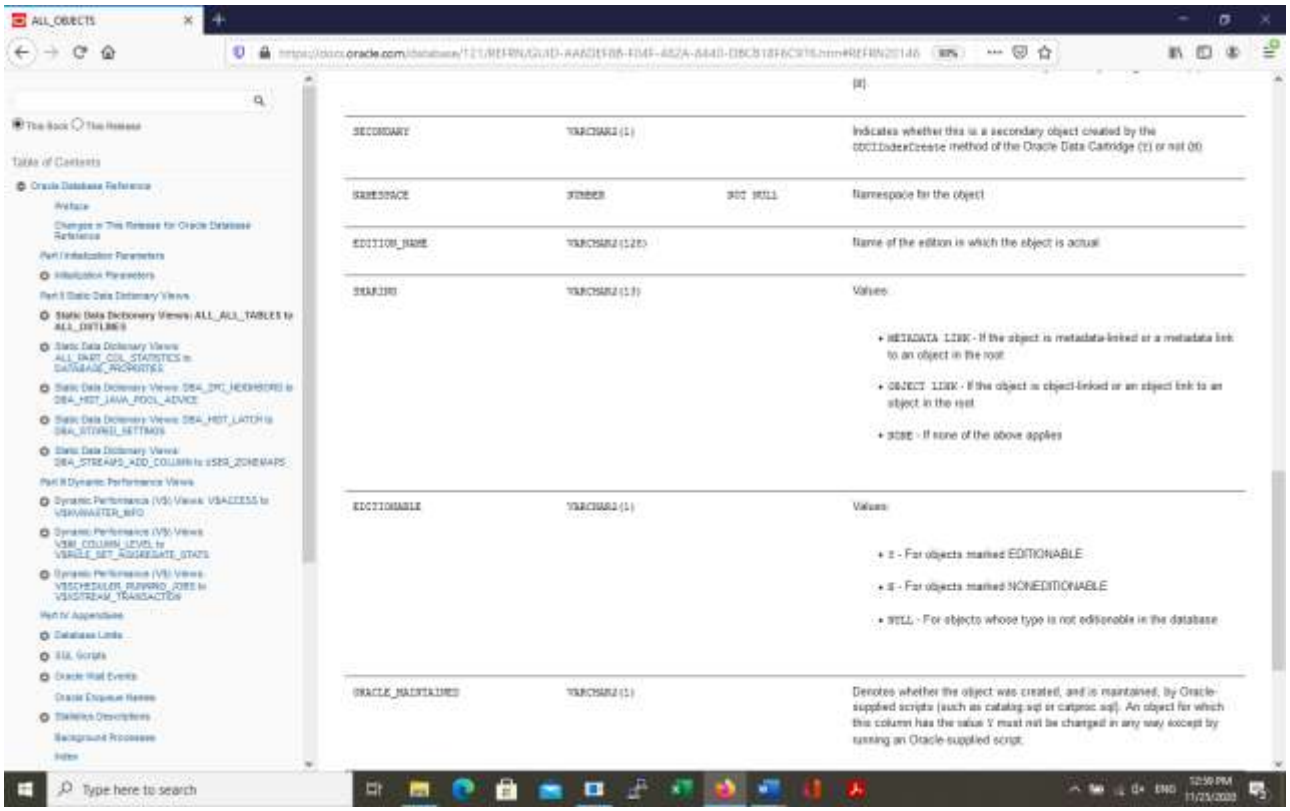
```
SQL> select table_name, partition_name, high_value from user_tab_partitions
        where table_name = 'ANGAJATI_DEP'
         order by partition_name;

TABLE_NAME                PARTITION_NAME            HIGH_VALUE
------------------------- ------------------------- ----------------------
ANGAJATI_DEP              DEP10                      10
ANGAJATI_DEP              DEP20                      20
ANGAJATI_DEP              DEP30                      30
```

15) Informatii despre obiectele din baza de date:

```
SQL> desc dba_objects
 Nume                                                  Nul?     Type
 ----------------------------------------------------- -------- ------
 OWNER                                                          VARCHAR2(30)
 OBJECT_NAME                                                    VARCHAR2(128)
 SUBOBJECT_NAME                                                 VARCHAR2(30)
 OBJECT_ID                                                      NUMBER
 DATA_OBJECT_ID                                                 NUMBER
 OBJECT_TYPE                                                    VARCHAR2(18)
 CREATED                                                        DATE
 LAST_DDL_TIME                                                  DATE
 TIMESTAMP                                                      VARCHAR2(19)
 STATUS                                                         VARCHAR2(7)
 TEMPORARY                                                      VARCHAR2(1)
 GENERATED                                                      VARCHAR2(1)
 SECONDARY                                                      VARCHAR2(1)
 NAMESPACE                                                      NUMBER
 EDITION_NAME                                                   VARCHAR2(128)
 SHARING                                                        VARCHAR2(13)
 EDITIONABLE                                                    VARCHAR2(1)
 ORACLE_MAINTAINED                                              VARCHAR2(1)
 SECONDARY                                                      VARCHAR2(1)
```


ALL_OBJECTS
• DBA_OBJECTS describes all objects in the database.
• USER_OBJECTS describes all objects owned by the current user. This view does not display the OWNER column.
Column | Datatype | NULL | Description
OWNER | VARCHAR2(128) | NOT NULL | Owner of the object
OBJECT_NAME | VARCHAR2(128) | NOT NULL | Name of the object
SUBOBJECT_NAME | VARCHAR2(128) | | Name of the subobject (for example, partition)
OBJECT_ID | NUMBER | NOT NULL | Dictionary object number of the object
DATA_OBJECT_ID | NUMBER | | Dictionary object number of the segment that contains the object
Note: OBJECT_ID and DATA_OBJECT_ID display data dictionary metadata. Do not confuse these numbers with the unique 16-byte object identifier (object ID) that Oracle Database assigns to row objects in object tables in the system.
OBJECT_TYPE | VARCHAR2(23) | | Type of the object (such as TABLE, INDEX)
CREATED | DATE | NOT NULL | Timestamp for the creation of the object
LAST_DDL_TIME | DATE | NOT NULL | Timestamp for the last modification of the object and dependent objects resulting from a DDL statement (including grants and revokes)
TIMESTAMP | VARCHAR2(19) | | Timestamp for the specification of the object (character data)
STATUS | VARCHAR2(7) | | Status of the object:
• VALID


ALL_OBJECTS
SECONDARY | VARCHAR2(1) | Indicates whether this is a secondary object created by the ODCIIndexCreate method of the Oracle Data Cartridge (Y) or not (N)
NAMESPACE | NUMBER | NOT NULL | Namespace for the object
EDITION_NAME | VARCHAR2(128) | | Name of the edition in which the object is actual
SHARING | VARCHAR2(13) | | Values:
• METADATA LINK - If the object is metadata-linked or a metadata link to an object in the root
• OBJECT LINK - If the object is object-linked or an object link to an object in the root
• NONE - If none of the above applies
EDITIONABLE | VARCHAR2(1) | | Values:
• Y - For objects marked EDITIONABLE
• N - For objects marked NONEDITIONABLE
• NULL - For objects whose type is not editionable in the database
ORACLE_MAINTAINED | VARCHAR2(1) | | Denotes whether the object was created, and is maintained, by Oracle-supplied scripts (such as catalog.sql or catproc.sql). An object for which this column has the value Y must not be changed in any way except by running an Oracle-supplied script.

```
SQL> select object_name, created
        from dba_objects
        where object_name like 'EMP%' and owner ='SCOTT';

OBJECT_NAME          CREATED
-------------------- ----------
EMP                  07-MAY-2019
EMP_COPY             01-NOV-2019
```

16) Informatii despre obiectele create de userul current:

```
SQL> desc user_objects

Name                                                      Null?    Type
 ----------------------------------------------------------------- -------- ------
 OBJECT_NAME                                                       VARCHAR2(128)
 SUBOBJECT_NAME                                                    VARCHAR2(30)
 OBJECT_ID                                                         NUMBER
 DATA_OBJECT_ID                                                    NUMBER
 OBJECT_TYPE                                                       VARCHAR2(19)
 CREATED                                                           DATE
 LAST_DDL_TIME                                                     DATE
 TIMESTAMP                                                         VARCHAR2(19)
 STATUS                                                            VARCHAR2(7)
 TEMPORARY                                                         VARCHAR2(1)
 GENERATED                                                         VARCHAR2(1)
 SECONDARY                                                         VARCHAR2(1)
 NAMESPACE                                                         NUMBER
 EDITION_NAME
VARCHAR2(128)
 SHARING                                                           VARCHAR2(13)
 EDITIONABLE                                                       VARCHAR2(1)
 ORACLE_MAINTAINED                                                 VARCHAR2(1)
 SECONDARY                                                         VARCHAR2(1)
```

```
SQL>  select object_name, object_type,created, status from user_objects;

OBJECT_NAME          OBJECT_TYPE          CREATED           STATUS
-------------------- -------------------- ---------         -------
PK_DEPT              INDEX                17-APR-07         VALID
DEPT                 TABLE                17-APR-07         VALID
EMP                  TABLE                17-APR-07         VALID
PK_EMP               INDEX                17-APR-07         VALID
BONUS                TABLE                17-APR-07         VALID
SALGRADE             TABLE                17-APR-07         VALID
V_STOCURI2           VIEW                 13-MAR-15         INVALID
```

CG$DISCIPLINE	PACKAGE	04-MAR-14	VALID
CG$DISCIPLINE	PACKAGE BODY	04-MAR-14	VALID
CG$BIS_STUDENTI	TRIGGER	04-MAR-14	VALID

17) Stergerea datelor dintr-o tabela, cu posibilitate de rollback:

```
SQL> delete from  emp_copy where deptno=10;
```

18) Trunchierea unei tabele (stergerea datelor dintr-o tabela, cu eliberarea spatiului ocupat de tabela si fara posibilitate de rollback):

```
SQL> truncate table emp_copy;
```

19) Stergerea unei tabele din dictionar, cu stergerea tuturor referintelor pe tabela:

```
SQL> drop table emp_copy cascade constraints;
```

20) Stergerea unei linii dintr-o tabela care contine duplicate folosind ROWID:

```
SQL> create table emp_del as select * from emp;
Table created.

SQL> insert into emp_del  select * from emp;
14 rows created.

SQL> SELECT rowid, empno, ename FROM emp_del order by empno;
```

ROWID	EMPNO	ENAME
AAAQwSAAEAAAAKUAAA	7369	SMITH
AAAQwSAAEAAAAKWAAA	7369	SMITH
AAAQwSAAEAAAAKWAAB	7499	ALLEN
AAAQwSAAEAAAAKUAAB	7499	ALLEN
AAAQwSAAEAAAAKUAAC	7521	WARD
AAAQwSAAEAAAAKWAAC	7521	WARD
AAAQwSAAEAAAAKWAAD	7566	JONES
AAAQwSAAEAAAAKUAAD	7566	JONES
AAAQwSAAEAAAAKUAAE	7654	MARTIN
AAAQwSAAEAAAAKWAAE	7654	MARTIN

……..

```
SQL> delete from emp_del where rowid='AAAQwSAAEAAAAKUAAB'; (ALLEN)
1 row deleted.
```

SQL> select rowid, empno, ename from emp_del order by empno;

ROWID	EMPNO	ENAME
AAAQwSAAEAAAAKUAAA	7369	SMITH
AAAQwSAAEAAAAKWAAA	7369	SMITH
AAAQwSAAEAAAAWUAAB	7499	ALLEN
AAAQwSAAEAAAAKUAAC	7521	WARD
AAAQwSAAEAAAAKWAAC	7521	WARD
AAAQwSAAEAAAAKWAAD	7566	JONES
AAAQwSAAEAAAAKUAAD	7566	JONES
AAAQwSAAEAAAAKUAAE	7654	MARTIN
AAAQwSAAEAAAAKWAAE	7654	MARTIN

……

---

21) Stergerea unei linii dintr-o tabela care contine duplicate folosind ROWNUM:

SQL> select rownum, empno, ename from emp_del order by empno;

ROWNUM	EMPNO	ENAME
1	7369	SMITH
15	7369	SMITH
16	7499	ALLEN
2	7499	ALLEN
3	7521	WARD
17	7521	WARD
18	7566	JONES
4	7566	JONES

……

SQL> delete from emp_del where rownum=16;
0 rows deleted.

SQL> delete from emp_del where rownum>16 and rownum<18;
0 rows deleted.

SQL> delete from emp_del where rownum<15;
14 rows deleted.

SQL> select rownum, empno, ename from emp_del order by empno;

ROWNUM	EMPNO	ENAME
1	7369	SMITH

2	7499	ALLEN
3	7521	WARD
4	7566	JONES
5	7654	MARTIN
6	7698	BLAKE
7	7782	CLARK
8	7788	SCOTT
9	7839	KING
10	7844	TURNER
11	7876	ADAMS
12	7900	JAMES
13	7902	FORD
14	7934	MILLER

14 rows selected.

SQL> create table emp_copy2 as select * from emp; -- cream o noua copie
Table created.

SQL> insert into emp_copy2 select * from emp; -- duplicam liniile in tabela emp_copy2
14 rows created.

SQL> select rownum, empno, ename from emp_copy2 order by empno;

ROWNUM	EMPNO	ENAME
1	7369	SMITH
15	7369	SMITH
16	7499	ALLEN
2	7499	ALLEN
3	7521	WARD
17	7521	WARD
18	7566	JONES
4	7566	JONES
5	7654	MARTIN
19	7654	MARTIN
6	7698	BLAKE
20	7698	BLAKE
7	7782	CLARK
21	7782	CLARK
8	7788	SCOTT
22	7788	SCOTT
9	7839	KING
23	7839	KING
10	7844	TURNER
24	7844	TURNER
11	7876	ADAMS

```
        25       7876  ADAMS
        12       7900  JAMES
        26       7900  JAMES
        13       7902  FORD
        27       7902  FORD
        28       7934  MILLER
        14       7934  MILLER

28 rows selected.
```

SQL> delete from emp_copy2 where rownum=(select min(rownum) from emp_copy2 where empno=7566); (JONES)
1 row deleted.

SQL> select rownum, empno, ename from emp_copy2 order by empno;

```
ROWNUM    EMPNO  ENAME
----------  ----------  ----------
        14       7369   SMITH
         1       7499  ALLEN
        15       7499  ALLEN
         2       7521  WARD
        16       7521  WARD
        17       7566  JONES
         3       7566  JONES
         4       7654  MARTIN
        18       7654  MARTIN
         5       7698  BLAKE
        19       7698  BLAKE
         6       7782  CLARK
        20       7782  CLARK
         7       7788  SCOTT
        21       7788  SCOTT
         8       7839  KING
        22       7839  KING
         9       7844  TURNER
        23       7844  TURNER
        10       7876  ADAMS
        24       7876  ADAMS
        11       7900  JAMES
        25       7900  JAMES
        12       7902  FORD
        26       7902  FORD
        13       7934  MILLER
        27       7934  MILLER

27 rows selected.
```

Obs. In tabela copy2_dep a fost stearsa linia cu rownum=1 (SMITH) in loc de linia cu rownum=4 (JONES) !! Explicatia este ca subcererea lucreaza astfel:

```
SQL> select rownum from emp_copy2 where empno=7566;

   ROWNUM
----------
         1
         2

SQL>  select min(rownum) from emp_copy2 where empno=7566;

MIN(ROWNUM)
-----------
          1
```

Concluzii:

a) Pentru stergerea manuala a liniilor se poate folosi ROWID dar Oracle nu recomanda aceasta metoda.

b) Se poate folosi stergerea in bloc a liniilor duplicat folosind ROWNUM, in anumite cazuri particulare, dar cu mare atentie. Nici aceasta metoda nu este recomandata de catre Oracle.

c) Pentru stergerea liniilor duplicat se recomanda utilizarea executiilor ciclice din PL/SQL (LOOP, FOR, WHILE, cursoare, etc).

22) Stergerea **unei singure linii duplicat dintr-o tabela care contine duplicate,** folosind subcereri:

```
SQL> select * from emp_copy2 A where rowid
       IN (select min(rowid) from emp_copy2 B where ename='SMITH');

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM
DEPTNO
---------- ---------- --------- ---------- --------- ---------- ---------- ----------
      7369 SMITH      CLERK           7902 17-DEC-80        800                    20

SQL> delete from emp_copy2 A where rowid IN
       (select min(rowid) from emp_copy2 B where ename='WARD');

1 row deleted.
```

SQL> select rownum, empno, ename from emp_copy2 order by empno;

ROWNUM	EMPNO	ENAME
1	7369	SMITH
14	7369	SMITH
2	7499	ALLEN
15	7499	ALLEN
16	7521	WARD
3	7566	JONES
17	7566	JONES
4	7654	MARTIN
18	7654	MARTIN

23. Stergerea **liniilor duplicat dintr-o tabela care contine cate un duplicat la fiecare linie**, folosind subcereri:

```
SQL> select * from emp_copy2 A where
        rowid IN (select min(rowid) from emp_copy2 B where A.empno=B.empno) order
by ename;

SQL>  delete  from emp_copy2 A where
        rowid IN (select min(rowid) from emp_copy2 B where A.empno=B.empno);
```

14 rows deleted.

24. Stergerea **tuturor liniilor duplicat dintr-o tabela care contine mai multe duplicate** folosind subcereri:

```
SQL> select * from emp_copy2 A where
        rowid > (select min(rowid) from emp_copy2 B where A.empno=B.empno) order
by ename;

SQL> delete  from emp_copy2 A where
        rowid >(select min(rowid) from emp_copy2 B where A.empno=B.empno);
```

28 rows deleted.

SQL> select rownum, empno, ename from emp_copy2 order by empno;

ROWNUM	EMPNO	ENAME
1	7369	SMITH
2	7499	ALLEN
3	7521	WARD

4	7566	JONES
5	7654	MARTIN
6	7698	BLAKE
7	7782	CLARK
8	7788	SCOTT
9	7839	KING
10	7844	TURNER
11	7876	ADAMS
12	7900	JAMES
13	7902	FORD
14	7934	MILLER

14 rows selected.

Exercitii:

1. Faceti o lista cu toate tabelele create de userul curent, tablespace-ul in care sunt create si proprietarul lor.

2. Sa se creeze o tabela copie EMP_LAB a tabelei EMP din userul curent. Dezactivati coloanele salariu si comision din tabela copie si verificati in dictionar cu cate coloane dezactivate figureaza tabela copie.

3. Sa se creeze linii duplicat pentru toti angajatii in tabela EMP_LAB. Sa se stearga apoi liniile duplicat din aceasta tabela.

4. Faceti o lista cu numele instantei curente, numele tablespace-lui permanent asignat userului curent, numele tabelelor create de userul curent si numarul maxim de extensii permise pentru fiecare tabela.

5. Sa se creeze un view DEPT_VIEW pe tabela DEPT din userul curent apoi sa se arate din dictionar proprietarul, numele, tipul si data cand a fost creat.